Category Theory for Mathematical Physics — Exercise Sheet 12

Morita equivalence and the Drinfeld centre

Variant: Solutions

Conventions as in Lecture 12: a half-braiding is $\gamma_X : X \otimes Z \to Z \otimes X$, and the braiding of $\mathcal{Z}(C)$ is $c_{(Z,\gamma),(Z',\gamma')} = \gamma'_Z$ (Theorem 12.7). The toric code's anyons are $1 = (0,1)$, $e = (0,\chi_-)$, $m = (1,1)$, $\varepsilon = (1,\chi_-)$ in $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ (Theorem 12.8), with $\mathbb{Z}_2 = \{0,1\}$ written additively and $\chi_-(h) = (-1)^h$; here e is an anyon, and the identity of S_3 is written 1. $\zeta = e^{2\pi i/3}$, $\varphi = (1+\sqrt{5})/2$, and ω is a 3-cocycle (Exercise 6). S, T and the central charge c are those of Lecture 10.

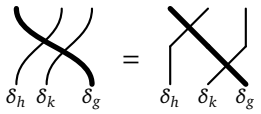
Exercise 1 (The S-matrix of the toric code from half-braidings). (i) Draw the condition of Definition 12.6 for a half-braiding on $Z = \delta_g$ in $\mathrm{Vec}_{\mathbb{Z}_2}$, with $X = \delta_h$ and $Y = \delta_k$, as in Equation (12.1). Writing $\gamma_{\delta_h} = \gamma(h)\,\mathrm{id}$, show that γ is a character of $\mathbb{Z}_2$, and recover the four simple objects (g,χ) of Theorem 12.8.

(ii) Draw $c_{m,e}$ as in Equation (12.2), and show that the braiding is $c_{X,Y} = \chi_Y(g_X)$. Write the table of $c_{X,Y}$ and of the double braidings $c_{Y,X}c_{X,Y}$ for $X, Y \in \{1, e, m, \varepsilon\}$.

(iii) Using that every d is 1 and every anyon is its own dual, show that $\tilde{S}_{XY}$ is the double braiding, and compare $S = \tilde{S}/2$ with Example 10.14.

(iv) Compute T and the central charge.

Solution. (i) The thick strand $Z = \delta_g$ is crossed by $\delta_h \otimes \delta_k$ at once (left), or by δ_k and then by δ_h (right):



Every object here is one-dimensional and every associator is 1, so each crossing is a scalar and the picture reads $\gamma(h+k) = \gamma(h)\gamma(k)$; at $h = k = 0$ it gives $\gamma(0) = \gamma(0)^2$, so $\gamma(0) = 1$. So γ is a homomorphism $\mathbb{Z}_2 \to \mathbb{C}^\times$, a character: $\chi_+ = 1$ or χ_-. With $g \in \{0,1\}$ this gives the four objects (g,χ) of Theorem 12.8: $1, e, m, \varepsilon$ (an object with a half-braiding is a sum of these, degree by degree).

(ii) By Theorem 12.7, $c_{X,Y}$ is the strand X crossing the thick strand Y by Y's half-braiding:

$$c_{m,e} \;=\; \underset{m\quad e}{\times} \;=\; \gamma_e(\delta_1) = \chi_-(1) = -1.$$

In general $c_{X,Y} = \gamma_Y(\delta_{g_X}) = \chi_Y(g_X)$: rows X with $g_X = 0$ (1 and e) are all 1; rows with $g_X = 1$ (m and ε) read $\chi_Y(1) = 1, -1, 1, -1$ for $Y = 1, e, m, \varepsilon$. The double braiding $c_{Y,X}c_{X,Y}$ is the table times its transpose entrywise: it is -1 for $\{e,m\}$, $\{e,\varepsilon\}$, $\{m,\varepsilon\}$ and 1 otherwise (including $X = Y$).

(iii) $\tilde{S}_{XY}$ is the Hopf link of X and $\bar{Y} = Y$ (Definition 10.2), $\mathrm{Tr}(c_{Y,X}c_{X,Y})$; for invertible objects the double braiding is a scalar on the line $X \otimes Y$, and the trace multiplies it by $d_{X\otimes Y} = 1$. So $\tilde{S}$ is the double-braiding table: rows $(1,1,1,1)$, $(1,1,-1,-1)$, $(1,-1,1,-1)$, $(1,-1,-1,1)$, and with dim = 4, $S = \tilde{S}/2$ is the matrix of Example 10.14, now from half-braidings alone.

(iv) The spins are the self-braidings, $\theta_X = c_{X,X} = \chi_X(g_X)$: $1, 1, 1, -1$, so $T = \mathrm{diag}(1,1,1,-1)$. The Gauss sum is $1+1+1-1 = 2 = \sqrt{4}\,e^0$: $c = 0$, as for every centre of a spherical category (Lecture 12).

Exercise 2 (The double $D(G)$, and $G=\mathbb{Z}_3$). (i) Show that $D(G)$ (Theorem 12.9) is associative, with unit $\sum_g \delta_g \otimes 1$, and has dimension $|G|^2$.

(ii) Let G be abelian. Show that $D(G)$ is commutative, and that for $a \in G$ and a character χ of G, $\rho_{a,\chi}(\delta_g \otimes h) = \delta_{a,g}\,\chi(h)$ is an algebra map $D(G) \to \mathbb{C}$.

(iii) For $G=\mathbb{Z}_3$, list the nine irreducible representations and their spins $\theta = \chi(a)$, and compare with Theorem 12.8.

(iv) Label the simple objects of $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_3})$ by (a,b), with flux a and character $\chi_b(h) = \zeta^{bh}$. Using Definition 10.2 and the braiding of Theorem 12.8, show that $S_{(a,b),(a',b')} = \zeta^{-(ab'+a'b)}/3$. For invertible S and T, $(ST)^3 = S^2$ is equivalent to $STS = T^{-1}ST^{-1}$: check the entry at $X=(0,1)$, $Y=(1,0)$, and show that it fails there for the complex conjugate $\bar{S}$ with the same T.

Solution. (i) The two orders give

$$\begin{aligned}\big((\delta_g\otimes h)(\delta_{g'}\otimes h')\big)(\delta_{g''}\otimes h'') &= \delta_{g,hg'h^{-1}}\,(\delta_g\otimes hh')(\delta_{g''}\otimes h'')\\ &= \delta_{g,hg'h^{-1}}\,\delta_{g,hh'g''(hh')^{-1}}\,\delta_g\otimes hh'h'',\\ (\delta_g\otimes h)\big((\delta_{g'}\otimes h')(\delta_{g''}\otimes h'')\big) &= \delta_{g',h'g''h'^{-1}}\,(\delta_g\otimes h)(\delta_{g'}\otimes h'h'')\\ &= \delta_{g',h'g''h'^{-1}}\,\delta_{g,hg'h^{-1}}\,\delta_g\otimes hh'h''.\end{aligned}$$

Given $g = hg'h^{-1}$, the conditions $g' = h'g''h'^{-1}$ and $g = hh'g''h'^{-1}h^{-1}$ are equivalent, so the two agree. The unit: $(\sum_{g'}\delta_{g'}\otimes 1)(\delta_g\otimes h) = \delta_g\otimes h$ (only $g'=g$ survives) and $(\delta_g\otimes h)(\sum_{g'}\delta_{g'}\otimes 1) = \sum_{g'}\delta_{g,hg'h^{-1}}\delta_g\otimes h = \delta_g\otimes h$ (only $g' = h^{-1}gh$). The basis $\delta_g\otimes h$ has $|G|^2$ elements.

(ii) For abelian G, $\delta_{g,hg'h^{-1}} = \delta_{g,g'}$, so the product $\delta_{g,g'}\,\delta_g\otimes hh'$ is symmetric in the two factors. And $\rho_{a,\chi}$ of the product is $\delta_{g,g'}\delta_{a,g}\chi(hh') = \delta_{a,g}\chi(h)\cdot\delta_{a,g'}\chi(h')$, the product of the values; it sends the unit to $\sum_g\delta_{a,g}=1$.

(iii) An irreducible representation of the commutative algebra $D(\mathbb{Z}_3)$ is one-dimensional by Schur's lemma, so it is an algebra map $D(\mathbb{Z}_3)\to\mathbb{C}$. Distinct algebra maps are linearly independent, so there are at most $\dim D(\mathbb{Z}_3) = 9$. For $\mathbb{Z}_3$ the characters are $\chi_b(h)=\zeta^{bh}$, $b\in\mathbb{Z}_3$, giving nine maps $\rho_{a,b}$; they are distinct, so they are all the irreducible representations. By Theorem 12.9 the spin is $\chi_b(a)/1 = \zeta^{ab}$: equal to 1 when $a=0$ or $b=0$ (five labels), ζ for $(a,b)=(1,1),(2,2)$ and ζ^2 for $(1,2),(2,1)$. These are the labels (g,χ) of Theorem 12.8 for $A=\mathbb{Z}_3$, with the same spins $\chi(g)$, as $\mathrm{Rep}D(G)\simeq\mathcal{Z}(\mathrm{Vec}_G)$ requires.

(iv) The objects are invertible with $d=1$, and $\overline{(a',b')} = (-a',-b')$. As in Exercise 1, $\tilde{S}_{XY}$ is the double braiding of X and $\bar{Y}$, and $c_{X,Y} = \chi_{b_Y}(a_X)$, so $\tilde{S}_{XY} = \zeta^{-b'a}\zeta^{-ba'}$; with $\dim=9$, $S=\tilde{S}/3$. If $(ST)^3=S^2$, cancelling S on the left gives $TSTST=S$, and multiplying by T^{-1} on both sides gives $STS=T^{-1}ST^{-1}$; the steps reverse. At $X=(0,1)$, $Y=(1,0)$, $\theta_X=\theta_Y=1$ and the right side is $S_{XY}=\zeta^{-1}/3=\zeta^2/3$. With $Z=(a,b)$, $S_{XZ}=\zeta^{-a}/3$, $\theta_Z=\zeta^{ab}$ and $S_{ZY}=\zeta^{-b}/3$, so the left side is

$$\tfrac{1}{9}\sum_{a,b}\zeta^{ab-a-b} = \tfrac{1}{9}\zeta^{-1}\sum_{u,v}\zeta^{uv} = \tfrac{1}{3}\zeta^2,$$

with $u=a-1$, $v=b-1$ and $\sum_{u,v}\zeta^{uv}=3$ (only $u=0$ contributes). For $\bar{S}$ the right side is $\zeta/3$, and the left side is $\frac{1}{9}\sum_{a,b}\zeta^{ab+a+b} = \frac{1}{9}\zeta^{-1}\sum_{u,v}\zeta^{uv} = \zeta^2/3 \neq \zeta/3$. So the conjugate S fails $(ST)^3=S^2$ here, though it agrees with S for the self-dual toric code. (The script checks the whole relation: it holds for S and fails for $\bar{S}$.)

Exercise 3 (The eight anyons of $\mathcal{Z}(\mathrm{Vec}_{S_3})$). (i) List the conjugacy classes of S_3 and their centralisers, and the simple objects $([g],\pi)$ of Theorem 12.9.

(ii) The dimension of $([g],\pi)$ is $|[g]|\dim\pi$. Check that the squares add up to $|S_3|^2$.

(iii) Compute the spins $\theta = \chi_\pi(g)/\dim\pi$.

Solution. (i) The classes are $\{1\}$, the three transpositions and the two 3-cycles. Their centralisers are S_3; for the transposition (12), the subgroup $\{1,(12)\} \cong \mathbb{Z}_2$; and for (123), $\{1,(123),(132)\} \cong \mathbb{Z}_3$. The simple objects are $([g],\pi)$ with π an irreducible representation of the centraliser: three for $\{1\}$ ($\mathbf{1}$, sgn, std), two for the transpositions, three for the 3-cycles: $3+2+3=8$.

(ii) The object $([g],\pi)$ is $\bigoplus_{g'\in[g]} \delta_{g'} \otimes V_{g'}$ with each $V_{g'}$ of dimension $\dim\pi$ (Lecture 12), so its dimension is $|[g]|\dim\pi$: $1,1,2;\ 3,3;\ 2,2,2$. The squares add up to $1+1+4+9+9+4+4+4 = 36 = 6^2$, as $\dim\mathcal{Z}(\mathrm{Vec}_G) = |G|^2$ requires.

(iii) For $\{1\}$, $\chi_\pi(1)/\dim\pi = 1$: three bosons. For the transpositions, the two characters of $\mathbb{Z}_2$ at the generator: 1 and -1 (a fermion). For the 3-cycles, the three characters of $\mathbb{Z}_3$ at (123): $1,\zeta,\zeta^2$. So $\mathcal{Z}(\mathrm{Vec}_{S_3})$ has eight anyons, of spins $1,1,1,1,-1,1,\zeta,\zeta^2$ and dimensions $1,1,2,3,3,2,2,2$.

Exercise 4 (Rep($\mathbb{Z}_2$), and the symmetry $e \leftrightarrow m$). (i) Following the proof of Theorem 12.2(ii), find the simple module endofunctors (V,s) of $\mathrm{Vect}_{\mathrm{fd}}$ over $\mathrm{Vec}_{\mathbb{Z}_2}$ and their composition, and show that $(\mathrm{Vec}_{\mathbb{Z}_2})^*_{\mathrm{Vect}_{\mathrm{fd}}} \simeq \mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$.

(ii) Show that the exchange $e \leftrightarrow m$ (fixing 1 and ε) preserves the fusion rules, the spins and the double braidings of the toric code. Does it preserve each $c_{X,Y}$?

(iii) Write each anyon as $X = (g_X, b_X) \in \mathbb{Z}_2 \times \mathbb{Z}_2$ with $\chi = \chi_-^{b_X}$; the toric code has trivial associators. Let F be the exchange $(g,b) \mapsto (b,g)$ on simple objects, extended linearly, with $J_{X,Y} = (-1)^{g_X b_Y}$ and $J_0 = 1$. Show that (F,J,J_0) is a monoidal functor (Definition 3.10) and that it is *braided*, $F(c_{X,Y}) \circ J_{X,Y} = J_{Y,X} \circ c_{FX,FY}$: the exchange is a braided autoequivalence.

Solution. (i) A module endofunctor is a space V with $s_g \in \mathrm{GL}(V)$ and $s_{g+h} = s_h s_g$, so $g \mapsto s_g^{-1}$ is a representation of $\mathbb{Z}_2$; module natural transformations are intertwiners, and composition tensors the representations (Theorem 12.2). The simple ones are the irreducible representations: $V = \mathbb{C}$ with $s_1 = 1$ or $s_1 = -1$. They compose by multiplying the signs, so the trivial one is the unit and the sign squares to it: the fusion rules of $\mathbb{Z}_2$. The associator is that of $\mathrm{Vect}_{\mathrm{fd}}$, trivial, so $(\mathrm{Vec}_{\mathbb{Z}_2})^*_{\mathrm{Vect}_{\mathrm{fd}}} \simeq \mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$ ($\omega = 1$, Example 12.3).

(ii) Fusion: $e\otimes e = m\otimes m = 1$, $e\otimes m = m\otimes e = \varepsilon$, $e\otimes\varepsilon = m$, $m\otimes\varepsilon = e$: exchanging e and m maps this list to itself. Spins: $\theta_e = \theta_m = 1$. Double braidings (Exercise 1): -1 for every pair of distinct anyons among e, m, ε, a condition symmetric under the exchange. But the single braidings are not preserved: $c_{e,m} = 1$ while $c_{m,e} = -1$, so the exchange needs a nontrivial tensor structure, (iii).

(iii) F is a functor, and $F(X\otimes Y) = F(X)\otimes F(Y)$ on simple objects, as $(g,b) \mapsto (b,g)$ is an automorphism of $\mathbb{Z}_2\times\mathbb{Z}_2$. With trivial associators on both sides and every J a scalar, the condition of Definition 3.10 is $J_{X\otimes Y,Z}J_{X,Y} = J_{X,Y\otimes Z}J_{Y,Z}$, that is $(-1)^{(g_X+g_Y)b_Z+g_Xb_Y} = (-1)^{g_X(b_Y+b_Z)+g_Yb_Z}$, true; the unit conditions hold as $J_{1,X} = J_{X,1} = 1$. The braiding is $c_{X,Y} = \chi_-^{b_Y}(g_X) = (-1)^{g_Xb_Y} = J_{X,Y}$, and F is the identity on scalars, so the left side of the braided condition is $(-1)^{2g_Xb_Y} = 1$ and the right side is $J_{Y,X}\,c_{FX,FY} = (-1)^{g_Yb_X}(-1)^{b_Xg_Y} = 1$. So F is a braided autoequivalence, the one Lecture 12 cites from Kong–Zhang §4.3.2: the single braidings are not invariant, only the double braidings and spins. Lecture 12 calls it the shadow of (i): $\mathrm{Vec}_{\mathbb{Z}_2}$ has two indecomposable module categories, the regular one with dual $\mathrm{Vec}_{\mathbb{Z}_2}$ and $\mathrm{Vect}_{\mathrm{fd}}$ with dual $\mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$, and the two ways of being self-dual exchange the charge e and the flux m.

Exercise 5 (Centres of modular categories). You may use that on $\mathcal{C}\boxtimes\mathcal{D}$ the braiding, the dimensions and the spins are taken factor by factor, $d_{a\boxtimes b} = d_a d_b$ and $\theta_{a\boxtimes b} = \theta_a\theta_b$, and that $\mathcal{C}^{\mathrm{rev}}$ has the spins $\bar\theta_a$.

(i) By Theorem 12.10(iii), $\mathcal{Z}(\mathrm{Fib}) \simeq \mathrm{Fib}\boxtimes\mathrm{Fib}^{\mathrm{rev}}$. List its simple objects with their dimensions and spins, and check $\dim\mathcal{Z}(\mathrm{Fib}) = (1+\varphi^2)^2$.

(ii) Show that for any unitary modular C the Gauss sum of $C \boxtimes C^{\mathrm{rev}}$ is $\dim C$, so $c = 0$.
(iii) How many simple objects does $\mathcal{Z}(\mathrm{SU}(2)_k)$ have, and what is its dimension?

Solution. (i) The simple objects of $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ are $a \boxtimes b$ (Definition 12.4): $1 \boxtimes 1$, $\tau \boxtimes 1$, $1 \boxtimes \tau$, $\tau \boxtimes \tau$, with $d_{a\boxtimes b} = d_a d_b$: $1, \varphi, \varphi, \varphi^2$, and spins $\theta_a \bar{\theta}_b$: $1, e^{4\pi i/5}, e^{-4\pi i/5}, 1$. The dimension is $1 + \varphi^2 + \varphi^2 + \varphi^4 = (1+\varphi^2)^2 = (2+\varphi)^2$, which is $(\dim \mathrm{Fib})^2$, as Theorem 12.10(i) requires.

(ii) The Gauss sum of $C \boxtimes C^{\mathrm{rev}}$ is $\sum_{a,b} d_a^2 d_b^2 \theta_a \bar{\theta}_b = \big(\sum_a d_a^2 \theta_a\big)\overline{\big(\sum_b d_b^2 \theta_b\big)} = |p_+|^2$, using $d_b > 0$. By Theorem 10.9, $|p_+|^2 = \dim C = \sqrt{\dim(C \boxtimes C^{\mathrm{rev}})}$, real and positive, so $e^{2\pi i c/8} = 1$: $c = 0$. For Fibonacci the two halves have $c = \frac{14}{5}$ and $-\frac{14}{5}$.

(iii) $\mathrm{SU}(2)_k$ is modular (Lecture 10), so $\mathcal{Z}(\mathrm{SU}(2)_k) \simeq \mathrm{SU}(2)_k \boxtimes \mathrm{SU}(2)_k^{\mathrm{rev}}$, with $(k+1)^2$ simple objects $j \boxtimes j'$ and dimension $(\dim \mathrm{SU}(2)_k)^2 = \big((k+2)/(2\sin^2(\pi/(k+2)))\big)^2$ (Sheet 5); its central charge is 0 by (ii).

Exercise 6 ⋆. *The double semion.* Let $C = \mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$, $\mathbb{Z}_2 = \{0,1\}$, with $\omega(1,1,1) = -1$, the other values 1.

(i) For a half-braiding on δ_g, write $\gamma_{\delta_h} = \gamma(h)\,\mathrm{id}$ with $\gamma(0) = 1$. Show that Definition 12.6 becomes

$$\gamma(h+k) = \frac{\omega(g,h,k)\,\omega(h,k,g)}{\omega(h,g,k)}\,\gamma(h)\gamma(k).$$

(ii) Solve it: find the four simple objects of $\mathcal{Z}(C)$ and their spins.
(iii) Conclude with Theorem 12.11 that $\mathrm{Vec}_{\mathbb{Z}_2}$ and C are not Morita equivalent.

Solution. (i) Take $Z = \delta_g$, $X = \delta_h$, $Y = \delta_k$. All objects are simple and one-dimensional, so each arrow in the condition of Definition 12.6 is a scalar times the identity: $\alpha_{Z,X,Y} = \omega(g,h,k)$, $\alpha^{-1}_{X,Z,Y} = \omega(h,g,k)^{-1}$, $\alpha_{X,Y,Z} = \omega(h,k,g)$, $\gamma_X = \gamma(h)$, $\gamma_Y = \gamma(k)$, and $\gamma_{X\otimes Y} = \gamma(h+k)$. The condition $\gamma_{X\otimes Y} = \alpha_{Z,X,Y}(\gamma_X \otimes \mathrm{id})\alpha^{-1}_{X,Z,Y}(\mathrm{id} \otimes \gamma_Y)\alpha_{X,Y,Z}$ is the displayed equation. (A morphism between the two candidates δ_g with different γ is zero, so these are the simple objects if γ ranges over the solutions: every object of C is a sum of δ_g's, and a half-braiding on a sum restricts to its summands of each degree, as in Theorem 12.8.)

(ii) If h or k is 0 the equation is trivial (ω is normalised, $\gamma(0) = 1$). At $h = k = 1$, $\gamma(0) = 1$: $1 = \frac{\omega(g,1,1)\omega(1,1,g)}{\omega(1,g,1)}\gamma(1)^2$. For $g = 0$ the three ω's are 1: $\gamma(1)^2 = 1$, $\gamma(1) = \pm 1$. For $g = 1$ all three are $\omega(1,1,1) = -1$, so the factor is $(-1)(-1)/(-1) = -1$: $\gamma(1)^2 = -1$, $\gamma(1) = \pm i$. So there are four simple objects, $(\delta_0, \pm 1)$ and $(\delta_1, \pm i)$. Each is invertible with $d = 1$, and its spin is its self-braiding $c_{Z,Z} = \gamma_Z = \gamma(g)$ (Theorem 12.7): 1 and 1 for $g = 0$, i and $-i$ for $g = 1$. This is the *double semion*: (δ_1, i) is a semion, $(\delta_1, -i)$ an antisemion.

(iii) For an invertible object Z, $Z \otimes Z$ is simple and the self-braiding $c_{Z,Z}$ is a number; a braided equivalence carries simple invertible objects to simple invertible objects and preserves these numbers. The centre of $\mathrm{Vec}_{\mathbb{Z}_2}$, the toric code, has spins $1, 1, 1, -1$, while $\mathcal{Z}(C)$ has $1, 1, i, -i$: no bijection matches them, so the centres are not braided equivalent, and by Theorem 12.11 $\mathrm{Vec}_{\mathbb{Z}_2}$ and $\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$ are not Morita equivalent, as Lecture 12 states. The associator alone, invisible in the fusion rules, decides the bulk phase.